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1. Users, Roles & Access

users

Purpose: Every human account in the system (admins, technicians, restaurant managers, authorities).

Key fields:

- id – PK
- email – login username, unique
- password_hash – hashed password
- full_name, phone, locale
- is_active – to disable accounts without deleting
- created_at, updated_at

Relationships:

- 1:N with restaurant_user_access (a user can be associated with many restaurants).
- 1:N with work_orders as reported_by_user_id, created_by_user_id, assigned_to_user_id.
- 1:N with tickets (creator, assignee).
- 1:N with ticket_messages (sender).
- 1:N with comments (author).
- 1:N with documents / photos (uploaded_by_user_id).
- M:N with roles via user_roles.

roles

Purpose: Define global role types: ADMIN, TECHNICIAN, CUSTOMER, AUTHORITY.

Key fields:

- id – PK
- code – e.g. "ADMIN", "TECHNICIAN"
- name, description

Relationships:

- M:N with users through user_roles.

user_roles

Purpose: Link table connecting users to global roles.

Key fields:

- Composite PK: (user_id, role_id)

Relationships:

- N:1 to users.
- N:1 to roles.

restaurant_user_access

Purpose: Defines which user has what role in a specific restaurant (e.g. Manager at restaurant, Technician assigned to that site, Inspector, etc.).

Key fields:

- id – PK
- restaurant_id – FK → restaurants

- user_id – FK → users
- role_at_site – e.g. 'OWNER', 'MANAGER', 'TECHNICIAN', 'AUTHORITY', 'VIEW_ONLY'
- is_primary – mark main contact
- created_at

Relationships:

- N:1 to restaurants.
- N:1 to users.

Cardinality summary:

- A **user** can be linked to many **restaurants**.
- A **restaurant** can have many **users** linked (owners, managers, technicians, authorities).

2. Customers, Restaurants & Sections

customers

Purpose: Legal/contractual entity that pays: a restaurant group, chain, or single owner.

Key fields:

- id – PK
- name – e.g. “Bam Bam Restaurants Group”
- tax_id – CIF/NIF
- Billing address, email, phone, notes, timestamps.

Relationships:

- 1:N with restaurants.
- 1:N with invoices.

restaurants

Purpose: Physical site/location where work is done.

Key fields:

- id – PK
- customer_id – FK → customers
- name – e.g. “Bam Bam Glòries”
- code – internal site code
- Address fields, timezone, is_active, notes, timestamps.

Relationships:

- N:1 to customers.
- 1:N to restaurant_sections.
- 1:N to equipment.
- 1:N to maintenance_assignments.
- 1:N to work_orders.
- 1:N to documents, photos, tickets, non_conformities.
- 1:N to restaurant_user_access.
- 1:N to iot_devices.
- 1:N to invoices.

restaurant_sections

Purpose: Sub-areas within a restaurant: “Main Kitchen”, “Cold Kitchen”, “Bar”, “Basement”, etc.

Key fields:

- id – PK
- restaurant_id – FK → restaurants
- name, description, sort_order

Relationships:

- N:1 to restaurants.
- 1:N to equipment.
- 1:N to maintenance_assignments.
- 1:N to work_orders.

Usage example:

A work order is raised for Section “Kitchen” at Bam Bam Glòries even if no specific equipment is selected.

3. Equipment & Categories

equipment_categories

Purpose: High-level type of equipment used to group and assign appropriate PM plans.

Key fields:

- id – PK
- code – e.g. "REFRIGERATION", "COOKING", "EXTRACTION", "HVAC", "ELECTRICAL", "PLUMBING", "FIRE"
- name, description

Relationships:

- 1:N with equipment_models.
- 1:N with equipment.
- 1:N with maintenance_plan_templates.

equipment_models

Purpose: Generic model info used by multiple pieces of equipment.

Key fields:

- id – PK
- category_id – FK → equipment_categories
- manufacturer, model_number, description, power_rating_kw, notes.

Relationships:

- N:1 to equipment_categories.
- 1:N to equipment.

equipment

Purpose: Each physical asset: specific fridge, combi oven, hood fan, AC split, etc.

Key fields:

- id – PK
- restaurant_id – FK → restaurants

- section_id – FK → restaurant_sections (optional)
- category_id – FK → equipment_categories
- model_id – FK → equipment_models
- name – human label (“Walk-in Freezer”, “Hood Fan #2”)
- asset_tag – QR code / internal ID
- serial_number, location_note, install_date, warranty_expiry
- status – 'IN_SERVICE', 'OUT_OF_SERVICE', 'RETIRED'
- criticality – 'LOW', 'MEDIUM', 'HIGH'
- custom_fields – JSON for extras.

Relationships:

- N:1 to restaurants, restaurant_sections, equipment_models, equipment_categories.
- 1:N to maintenance_assignments.
- 1:N to work_orders.
- 1:N to non_conformities.
- 1:N to documents, photos.
- 1:N to iot_devices.

Workflow:

Equipment is the **anchor** for:

- IoT sensors,
- Preventive maintenance assignments,
- Corrective work orders,
- Certificates,
- Non-conformities.

4. Preventive Maintenance Plans & Assignments

maintenance_plan_templates

Purpose: Reusable maintenance templates (checklists) for categories or types of assets.

Key fields:

- id – PK
- name – e.g. “Monthly Extraction PM – Standard Kitchen”
- description
- category_id – FK → equipment_categories (optional)
- applies_to – 'EQUIPMENT' | 'RESTAURANT' | 'SECTION'
- frequency_type – 'WEEKLY', 'MONTHLY', 'QUARTERLY', 'SEMIANNUAL', 'ANNUAL', 'CUSTOM'
- frequency_days – used when 'CUSTOM'
- default_duration_min
- is_active
- created_by_user_id – who defined it (admin/engineer)

Relationships:

- N:1 to equipment_categories.
- 1:N to maintenance_plan_tasks.
- 1:N to maintenance_assignments.

maintenance_plan_tasks

Purpose: Lines in a PM template – each is one checklist item.

Key fields:

- id – PK
- plan_id – FK → maintenance_plan_templates
- section – logical grouping (“Extraction & Ventilation”, “Refrigeration”)
- task_order – display order
- description – e.g. “Check and clean hood filters”
- expected_result_type – 'BOOLEAN', 'NUMERIC', 'TEXT', 'CHOICE'
- unit, min_value, max_value, options_json
- is_mandatory

Relationships:

- N:1 to maintenance_plan_templates.
- 1:N to work_order_tasks (a WO task may reference one plan task).

maintenance_assignments

Purpose: Attaches a PM plan to a specific restaurant, section, or equipment with actual dates.

Key fields:

- id – PK
- plan_id – FK → maintenance_plan_templates
- restaurant_id – FK → restaurants (always set)
- section_id – FK → restaurant_sections (optional)
- equipment_id – FK → equipment (optional)
- active – enable/disable PM
- start_date, next_due_date, last_completed_date
- notes, timestamps

Relationships:

- N:1 to maintenance_plan_templates.
- N:1 to restaurants.
- N:1 to restaurant_sections (optional).
- N:1 to equipment (optional).
- 1:N to work_orders (each time the assignment becomes due, you create a WO).

Cardinality conceptually:

- One **plan template** → many **assignments** (one per site/asset).
- One **assignment** → many **work orders over time** (each period a visit).

5. Work Orders, Tasks & Non-conformities

work_orders

Purpose: Central operational entity: a maintenance intervention (preventive, corrective, inspection).

Key fields:

- id – PK
- restaurant_id – FK → restaurants (**mandatory**)
- section_id – FK → restaurant_sections (optional)

- equipment_id – FK → equipment (optional)
- assignment_id – FK → maintenance_assignments (optional – if generated from PM)
- type – 'PREVENTIVE','CORRECTIVE','INSPECTION'
- status – 'OPEN','IN_PROGRESS','COMPLETED','CANCELLED'
- priority – 'LOW','MEDIUM','HIGH','CRITICAL'
- summary, description
- reported_by_user_id – who raised the issue (customer/manager)
- created_by_user_id – who created WO in the system (admin/dispatcher)
- assigned_to_user_id – technician responsible
- scheduled_start, scheduled_end
- started_at, completed_at, cancelled_at, cancellation_reason
- customer_visible – hide internal WOs if needed
- timestamps

Relationships:

- N:1 to restaurants, restaurant_sections, equipment, maintenance_assignments.
- N:1 to users (reported_by, created_by, assigned_to).
- 1:N to work_order_tasks.
- 1:N to non_conformities.
- 1:N to photos.
- 1:N to tickets (or 1:1 link via tickets.related_work_order_id).
- 1:N to sensor_alerts (IoT → WO).
- 1:N to invoices.

work_order_tasks

Purpose: Actual checklist items completed by the technician during a WO.

Key fields:

- id – PK
- work_order_id – FK → work_orders
- plan_task_id – FK → maintenance_plan_tasks (optional; null for ad-hoc tasks)
- section – textual section name
- task_order – order in WO
- description – copied from template, editable
- Result fields:
 - result_boolean
 - result_numeric + unit
 - result_text
- is_non_conformity – if this answer indicates an issue
- timestamps

Relationships:

- N:1 to work_orders.
- N:1 to maintenance_plan_tasks.

non_conformities

Purpose: Explicit record of issues found during maintenance or inspection (for compliance and follow-up).

Key fields:

- id – PK
- work_order_id – FK → work_orders
- work_order_task_id – FK → work_order_tasks (optional)
- restaurant_id – FK → restaurants
- equipment_id – FK → equipment (optional)
- severity – 'LOW','MEDIUM','HIGH','CRITICAL'
- description, recommendation
- status – 'OPEN','IN_PROGRESS','RESOLVED'
- created_at, resolved_at

Relationships:

- N:1 to work_orders.
- N:1 to work_order_tasks (the specific checklist item that failed).
- N:1 to restaurants.
- N:1 to equipment.

Typical flow:

1. Technician executes work_order_tasks.
2. Marks one as is_non_conformity = true OR enters an anomalous value.
3. System creates a non_conformities record linked to the WO & equipment.
4. Later we may create a corrective work order referencing that non-conformity (via description or a linking field we can add in the future).

6. Documents & Photos

documents

Purpose: Store certificates, legal docs, manuals, inspection reports, etc.

Key fields:

- id – PK
- restaurant_id – FK → restaurants
- equipment_id – FK → equipment (optional)
- uploaded_by_user_id – FK → users
- name – file name / friendly name
- doc_type – 'CERTIFICATE','MANUAL','REPORT','OTHER'
- storage_path – location in storage (S3 key, local path)
- issue_date, expiry_date, issuing_body
- notes, created_at

Relationships:

- N:1 to restaurants, equipment, users.

photos

Purpose: Visual evidence for maintenance actions, damage, before/after pictures.

Key fields:

- id – PK
- restaurant_id – FK → restaurants
- equipment_id – FK → equipment (optional)
- work_order_id – FK → work_orders (optional)
- uploaded_by_user_id – FK → users

- storage_path, caption, created_at

Relationships:

- N:1 to restaurants, equipment, work_orders, users.

7. Tickets, Messages & Comments

tickets

Purpose: Communication thread for issues/requests from customers, or internal coordination.

Key fields:

- id – PK
- restaurant_id – FK → restaurants
- created_by_user_id – FK → users
- assigned_to_user_id – FK → users (technician/dispatcher)
- type – 'ISSUE','REQUEST','QUESTION'
- status – 'OPEN','IN_PROGRESS','RESOLVED','CLOSED'
- priority – 'LOW','MEDIUM','HIGH','CRITICAL'
- subject
- related_work_order_id – FK → work_orders (optional)
- created_at, closed_at

Relationships:

- N:1 to restaurants.
- N:1 to users (creator, assignee).
- N:1 to work_orders (if a WO is created for the ticket).
- 1:N to ticket_messages.

ticket_messages

Purpose: Messages in a ticket (like comments in ACRON or helpdesk systems).

Key fields:

- id – PK
- ticket_id – FK → tickets
- sender_user_id – FK → users
- body – message text
- is_internal – true = visible only to technicians/admins
- created_at

Relationships:

- N:1 to tickets.
- N:1 to users.

comments

Purpose: Generic comments on various entities (work orders, equipment, invoices...).

Key fields:

- id – PK
- user_id – FK → users
- target_type –
'WORK_ORDER','EQUIPMENT','RESTAURANT','INVOICE','NON_CONFORMITY',
...

- target_id – ID in target table
- body, created_at

Relationships:

- N:1 to users.
- Polymorphic link to other entities (handled at app level).

8. (Optional Core) IoT: Devices & Sensors

iot_devices

- Linked to **one restaurant**, optionally a specific **equipment**.
- Describes gateway/device hardware: manufacturer, model, serial, device_type (gateway or node), status.

iot_sensors

- Linked to **one iot_device**.
- Describes each measurement channel: sensor_type (temperature, vibration), unit, allowed range.

sensor_readings

- Linked to **one iot_sensor**.
- Stores time series: ts, value_numeric, raw_payload.

sensor_alerts

- Linked to **one iot_sensor** and optionally a **work_order**.
- Represents threshold violations, offline device events, or anomaly detection that might auto-create a work order.